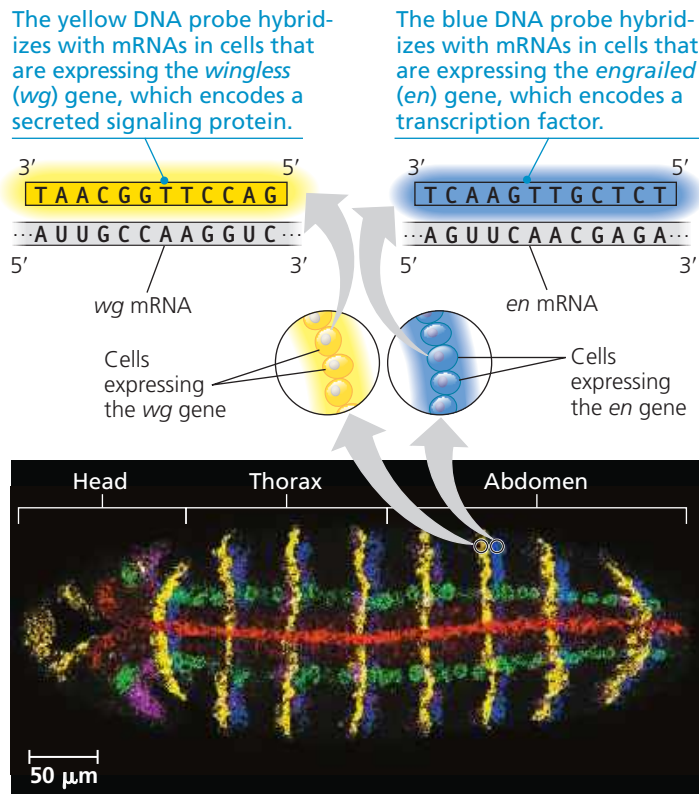


▼ **Figure 20.9 Determining where single genes are expressed by *in situ* hybridization analysis.** This *Drosophila* embryo was incubated in a solution containing DNA probes for five different mRNAs, each probe labeled with a different fluorescently colored tag. The embryo was then viewed using fluorescence microscopy. Each color marks where a specific gene is expressed as mRNA. The arrows from the groups of yellow and blue cells above the micrograph show a magnified view of nucleic acid hybridization of the appropriately colored probe to the mRNA. The thorax (trunk) and abdomen are made up of repeating segments. Yellow cells (expressing the *wg* gene) interact with blue cells (expressing the *en* gene); their interaction helps establish the pattern in a body segment.

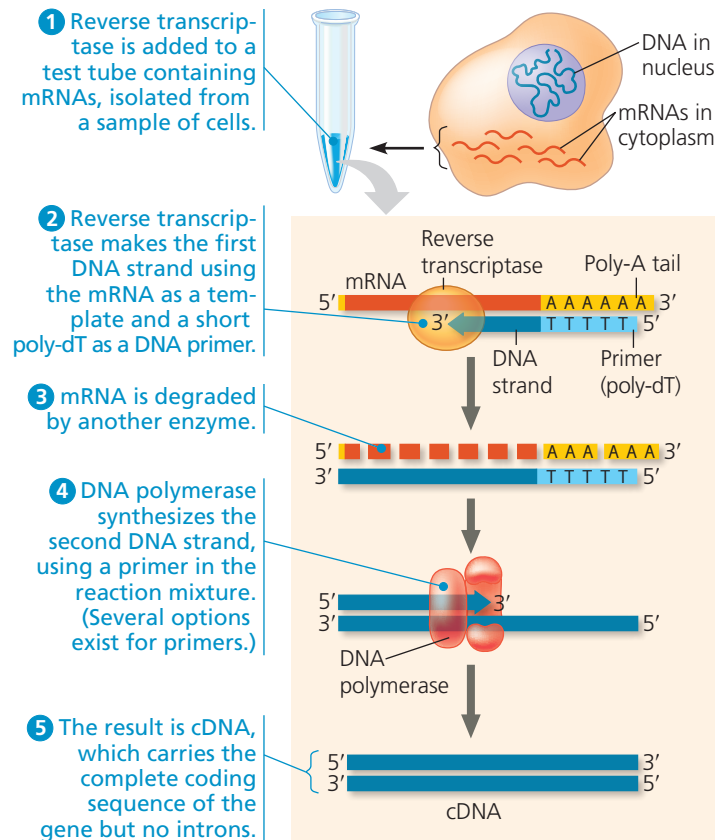


Different probes can be labeled with different fluorescent dyes, sometimes with strikingly beautiful results (Figure 20.9).

Other mRNA detection techniques may be preferable for comparing the amounts of a specific mRNA in several samples at the same time—for example, in different cell types or in embryos of different stages. One method that is widely used is called the **reverse transcriptase polymerase chain reaction**, or **RT-PCR**.

RT-PCR begins by turning sample sets of mRNAs into double-stranded DNAs with the corresponding sequences. First, the enzyme reverse transcriptase (from a retrovirus; see Figure 19.9) is used to synthesize a complementary DNA copy (a *reverse transcript*) of each mRNA in the sample (Figure 20.10). The mRNA is then degraded by addition of a specific enzyme, and a second DNA strand, complementary to the first, is synthesized by DNA polymerase. The resulting double-stranded DNA is called **complementary DNA**

▼ **Figure 20.10 Making complementary DNA (cDNA) from eukaryotic genes.** Complementary DNA is made in a test tube using mRNA as a template for the first strand. Only one mRNA is shown after step 1, but the final collection of cDNAs would reflect all the mRNAs present in the cell.



(**cDNA**). (Made from mRNA, cDNA lacks introns and can be used for protein expression in bacteria, as mentioned earlier.) To analyze the timing of expression of the *Drosophila* gene of interest, for example, we would first isolate all the mRNAs from different stages of *Drosophila* embryos and make cDNA from each stage (Figure 20.11).

Next in RT-PCR is the PCR step (see Figure 20.7). As you will recall, PCR is a way of rapidly making many copies of one specific stretch of double-stranded DNA, using primers that hybridize to the opposite ends of the segment that we are interested in studying. In our case, we would add primers corresponding to a segment of our *Drosophila* gene, using the cDNA from each embryonic stage as a template for PCR amplification in separate samples. When the products are analyzed on a gel, copies of the amplified region will be observed as bands only in samples that originally contained mRNA from the gene being studied. This method can tell researchers whether an mRNA is present. However, accurate measurement of mRNA levels requires newer, more quantitative PCR machines that use a dye that fluoresces only when bound to a double-stranded PCR product. This technique,